Report R9 — Sectors and basins: the weak-component decomposition over the rule space

Tier 1e, open boundaries

QCA Sector Analysis project (Tier 1)

2026-07-29 (`tier1e_version 1e.1`, `obc0`)

Abstract

The dissipative pipeline has so far reported terminal strongly connected components — "recurrent classes" — and their basins. Those are the right object for the *measured* circuit and the wrong object for a sector axis: they do not partition the basis, they are monitored-relative, and the Tier-2 wall grammar predicts sectors rather than attractors. This report adds the weakly connected component (WCC) as a first-class observable, which fixes all three at once: span(WCC) is an enclosure of the channel, invariant under every Kraus operator and therefore exact for the monitored and the unmonitored dynamics alike; WCCs partition the basis; and they are precisely the minimal projectors of the *diagonal* part of the commutant. We sweep all 256 rules at open boundaries, verify the sum rule and the cross-tier consistency assertions, and test the exclusion curve $ab \geq 2$ that a partition forces on the sector map. Two negative results are recorded prominently: the Tier-2 prediction $a_{\mathrm{wcc}} \geq \varphi$ for rule 28 is *false* — the sector count obeys an exact recurrence with base the plastic number $\rho = 1.32472$ — and the basin sum rule turned out to be unanswerable from the Tier-1a archive for 76 units, which we fixed by recomputing them independently. This report covers `obc0` only; `pbc` follows separately.

Contents

1 Three tiers, and why the WCC is the sector-level object **2**

1.1 Vocabulary . . . 2

1.2 Why terminal SCCs cannot carry a sector axis . . . 2

1.3 What a WCC is . . . 3

1.4 The algorithm, and one deliberate deviation . . . 3

2 The sweep **3**

3 Ground truths **4**

4 Rule 28: the wall-transparency prediction fails **4**

5 The hyperbola **5**

5.1 Anchors . . . 5

5.2 The check that carries the validation . . . 5

5.3 Verdicts in the base plane . . . 6

5.4 Where the rate estimate comes from . . . 6

5.5 Two bounds that the fitter needs and the theorem supplies . . . 7

6 Sectors against monitored attractors **7**

6.1 The sector axis is uninformative for the V+reset rules . . . 7

7 The corner view, and where the mass sits **10**
7.1 Corner plots, as in R2 fig. 6 10
7.2 Terminal SCCs against transient states 10
7.3 Correction (2026-08-01): α and the base came from different models 13

8 Clustering the dissipative rules by their coherent correspondent **14**
8.1 What adding dissipation does 15
8.2 Fragmentation in an open quantum system 15
8.3 Which constant describes what: a naming correction 17
8.4 The base plane hides a whole tier, and 150/105 are in it 17
8.5 A second polynomial mechanism: the pinned frontier 18

9 Rule 150's wall charge survives six dissipative rules **22**

10 Growth classes and exact bases **22**

11 Basins, and a gap in the Tier-1a archive **23**

12 Validation **24**

13 Comparison hooks for Tier 2 **25**

14 Open items **25**

15 Reproduce **25**

1 Three tiers, and why the WCC is the sector-level object

1.1 Vocabulary

The single most important thing this report changes is naming. Throughout the project from here on:

sector	weakly connected component; exact for *both* experiments
monitored attractor	terminal SCC; a property of the measured circuit
channel attractor	block of the fixed-point algebra (Tier 1b / 1d)

A terminal SCC is never a "sector"; a WCC is never an "attractor". Every table and figure caption below states which experiment its numbers describe.

1.2 Why terminal SCCs cannot carry a sector axis

(P1) They do not partition. Transient states belong to no recurrent class, so $\sum_k |A_k| \ll 2^N$ and no sum rule constrains the multiset. The unitary sectors *do* partition, so the existing figures put two different kinds of object on one axis. Rule 22 at $N = 12$ is the extreme case: three monitored attractors, each a single state, covering 3 of 4096 states, against two sectors covering all 4096.

(P2) They are monitored-relative. The transient/recurrent split is a property of the measured chain; the unmonitored channel can keep immortal weight on transient states (R5 rule 22, R7).

(P3) They are not what the analytics predicts. The Tier-2 wall grammar and transfer matrices give sector counts.

1.3 What a WCC is

Definition 1. The one-cycle support graph has nodes $x \in \{0,1\}^N$ and an edge $x \to y$ whenever $\langle y|\Phi_C(|x\rangle\langle x|)|y\rangle \neq 0$. A *sector* is a connected component of this graph read as undirected.

Proposition 1 (enclosure)**.** span(*sector*) *is invariant under every Kraus operator of* Φ_C*, hence invariant for the monitored and the unmonitored dynamics alike.*

The commutant reading is the one that connects this tier to R8. A diagonal operator D commutes with the dynamics iff $D_x = D_y$ whenever the support graph has an edge $x \to y$, i.e. iff D is constant on components. So

$$\dim\{\, D \text{ diagonal} : [D, \Phi_C] = 0 \,\} \;=\; \#\text{components} \;=\; n_{\text{wcc}}, \tag{1}$$

and the graph analysis computes *exactly* the diagonal part of the commutant — no more. That is why it cannot see reflection, or any other symmetry whose eigenvectors are superpositions rather than basis states (R8 §10.3: at $N = 11$ the diagonal part is 7-dimensional inside a commutant of dimension 23 360).

1.4 The algorithm, and one deliberate deviation

A single streaming pass over $\text{succ}(x)$ for $x = 0 \ldots 2^N - 1$, union-find with union by size and path compression, every edge symmetrised by construction ($\text{union}(x, y)$ for each $y \in \text{succ}(x)$). No adjacency storage, no DFS stack. This is the code path the unitary sweep already used, now factored into `graph/wcc.py::weak_components` and shared by both callers.

Deviation, and why. The Tarjan sweep aborts a unit as soon as one component exceeds $f_{\text{erg}} 2^N$. The WCC pass does *not*: aborting leaves an incomplete partition, which would break the sum rule A1 and destroy the exact anchors the exclusion-curve test needs — rule 51 has to come out as one sector of size exactly 2^N, not as "something exceeded half the space". Union-find is cheap enough to always finish, so `ergodic` here is a *classification* derived from $D_{\max}$, and the exponential saving lives in the sweep instead. A second guard: a rule is not written off as ergodic on the strength of a tiny chain, since at $N = 6$ almost anything connects past half of 64 states; the verdict must hold at two consecutive N and never below $N = 10$.

2 The sweep

All 256 rules at `obc0`, swept with N in the *outer* loop so that a run stopped against a wall-clock budget still leaves uniform coverage — every rule with the same N range and therefore comparable fits. Records live in their own store `results_tier1e/`, keyed on $(\text{rule}, N, \texttt{bc})$ and versioned `1e.1`.

A note on versioning. The task asks for the shared `ENGINE_VERSION` to be bumped to `1e.1`. Doing that would make `has_unit()` false for all ∼6000 archived Tier-1a records, so the next sweep would recompute the lot — which the same section explicitly forbids. The version bump therefore lives on the new tier, exactly as Tier 1d does. Nothing in Tier 1a/1b/1d is recomputed or invalidated.

N	rules covered	2^N
6	256	64
7	256	128
8	256	256
9	256	512
10	256	1 024
11	256	2 048
12	256	4 096
13	256	8 192
14	256	16 384
15	256	32 768
16	256	65 536
17	73	131 072
18	3	262 144

Coverage is uniform: all 256 rules at every $N = 6 \dots 16$, so the descriptors of §5 are all fitted on the same domain. A targeted set of 30 diagnostic rules was carried to $N = 17$–18 afterwards; the headline map deliberately does *not* use those extra points, since a rule fitted on a longer window is not comparable with one fitted on the uniform window.

3 Ground truths

All of the following are regression-tested in `tests/test_wcc.py`.

- **Rule 204** (`IIII`): 2^N sectors of size 1; $a = 2$, $b = 1$.
- **Rule 51** (`VVVV`): one sector of size 2^N; $a = 1$, $b = 2$.
- **Rule 150** `obc0`: sectors are the domain-wall-number eigenspaces, $|S_w| = \binom{N+1}{w}$ on even w. The computed values reproduce the reference series for $N = 8 \dots 17$ exactly — #sectors $5, 6, 6, 7, 7, 8, 8, 9, 9, 10$ and $D_{\max}$ 126, 210, 462, 924, 1716, 3003, 6435, 12870, 24310, 43758.
- Two of the task's regression targets are `pbc`-specified (rule 156's Lucas law and rule 22's attractor count). Both are implemented and pass in the test suite, but they are `pbc` statements and are therefore discussed in the `pbc` report, not here.

A naming trap worth recording. Wolfram numbers are not classical ECA rules in this encoding. W_{90} has symbol tuple `DEED` and is *dissipative*; assertion A2 must not be applied to it. Its $N = 8$ `obc0` numbers make the P1 point cleanly: sector sizes $112, 112, 16, 16$ summing to 256, against terminal-SCC sizes $7, 7, 1, 1$ summing to 16.

4 Rule 28: the wall-transparency prediction fails

The task predicts $a_{\text{wcc}} \geq \varphi$ for rule 28 (`IIVD`), on the grounds that it inherits the 01-wall grammar of rule 156, and asks for a prominent flag if that is violated. **It is violated.** The `obc0` sector count is

$$n_{\text{wcc}}(N) = 11, 15, 20, 27, 36, 48, 64, 85, 113, 150, 199 \qquad (N = 6 \dots 16),$$

which satisfies the exact integer recurrence

$$a_n = a_{n-1} + a_{n-2} - a_{n-4}, \tag{2}$$

verified out of sample on every available term. Its characteristic polynomial factors as $x^4 - x^3 - x^2 + 1 = (x-1)(x^3 - x - 1)$, so the growth base is the **plastic number**

$$\rho = 1.324717957\ldots \;<\; \varphi = 1.618033988\ldots,$$

the real root of $x^3 = x + 1$. This is an exact algebraic base, not a fit, so the gap to φ cannot be attributed to a short series or to a parity artifact.

What this falsifies is the *transparency* step of the argument, not the wall grammar itself: rule 28 replaces one of rule 156's symbols with a reset `D`, and the reset evidently removes wall configurations from circulation rather than merely relabelling them. Quantifying which configurations are lost is a Tier-2 question and is listed in §13.

5 The hyperbola

Theorem 1 (exclusion curve)**.** *Sectors partition the basis, so with $K \sim a^N$ sectors and $D_{\max} \sim b^N$,*

$$K\,D_{\max} \;\geq\; \sum_k D_k \;=\; 2^N \qquad \Longrightarrow \qquad ab \;\geq\; 2, \tag{3}$$

with equality iff the sector sizes are all comparable. No rule may lie below $b = 2/a$.

This is a theorem about *partitions*. It therefore constrains the sector map and says nothing whatever about the monitored-attractor map, where $a_{\text{att}} b_{\text{att}} < 2$ is expected and the deficit $2 - a_{\text{att}} b_{\text{att}}$ measures transient dominance (V4). We assert nothing there and report the deficit as an observable.

5.1 Anchors

rule	tuple	a want	b want	where	a	b	ab	verdict
204	`IIII`	2	1	on	2.0000	1.0000	2.0000	✓ on
51	`VVVV`	1	2	on	1.0000	2.0000	2.0000	✓ on
150	`IVVI`	1	2	on	1.0000	2.0000	2.0000	✓ on
156	`IIVI`	φ	$4^{1/5}$	above	1.6180	1.3195	2.1350	✓ above

Rules 204, 51 and 150 sit on the curve to six decimals, as they must, and rule 156 sits above it at exactly $\varphi \cdot 4^{1/5} = 2.1350$. How the rate behind that number is obtained — and why it must not be read off a BIC fit — is §5.4.

5.2 The check that carries the validation

The asymptotic statement $ab \geq 2$ is a corollary about *bases*, and it only means anything when both series are exponential: if the sector count is polynomial its base is 1 by convention, the product silently drops the polynomial factor, and the inequality degenerates to $b \geq 2$ approached from below at finite N. So the validation is carried instead by the theorem in its original, fit-free form,

$$n_{\text{wcc}}(N) \cdot D_{\max}(N) \;\geq\; 2^N \qquad \text{for every } N, \tag{4}$$

an immediate corollary of the A1 sum rule involving no fitting anywhere.

(4) holds for 256/256 **rules at every** N, and it is *tight*: the smallest value of $n_{\text{wcc}} D_{\max}/2^N$ anywhere in the sweep is exactly 1.0000 (rule 0 at $N = 6$).

5.3 Verdicts in the base plane

verdict	rules
on the curve, $ab = 2$	197
above, $ab > 2$	50
sub-exponential factor: ab test degenerate	7
irregular series, no growth law	2
below but inside the fit uncertainty	0
below, slack too wide to conclude	0
violation	0
total	256

The base-plane check is built so that a wide uncertainty band can never launder a sub-2 product into a silent pass. Four things are recorded separately: the raw product, whether it falls below 2 at all, whether the shortfall sits inside the propagated leave-one-out spread, and whether either series is sub-exponential so that the test is degenerate by construction. Every rule with $ab < 2$ is listed in full, as the task requires.

rule	tuple	a	b	ab	slack	$2/a$	fit b_{hi}
229	`EDIE`	1.0000	1.9188	1.9188	0.182	2.0000	2.1353
6	`IVDD`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
14	`IEDD`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
20	`IDVD`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
84	`IDED`	1.0000	1.9226	1.9226	0.090	2.0000	2.0493
74	`DEID`	1.0000	1.9322	1.9322	0.399	2.0000	2.1144
88	`DIED`	1.0000	1.9368	1.9368	0.226	2.0000	1.9777

There are no violations. Every remaining sub-2 rule (6, 14, 20, 74, 84, 88, 229) has a *polynomial* sector count — for instance W_{88} has $n_{\text{wcc}} = 3, 3, 4, 4, 4, 5, 5, 5, 6, 6, 6$, i.e. $\sim N/2$ — so $a = 1$, and the degenerate form of (3) demands $b \geq 2$, which a finite-N fit necessarily approaches from below (1.92–1.94 here). Their $D_{\max}/2^N$ decays slowly enough to be consistent with a power-law prefactor but not cleanly enough for the test of §5.4 to certify it; the theorem, not the fit, gives their base. Two further rules, W_{90} and W_{165}, have genuinely irregular series ($n_{\text{wcc}} = 2, 1, 4, 4, 2, 6, 2, 6, 12, 1, 20$) that no growth law describes; they are classified *irregular* and omitted from the base plane rather than placed at a fictitious coordinate.

5.4 Where the rate estimate comes from

R2 §3 already documented the trap that this sweep first walked into, and it is worth restating because it decides four of the numbers above. The BIC model $\ln y = c + \alpha \ln N + \kappa N$ is the right tool for the growth *class* and for the sub-leading power, but its $\alpha \ln N$ term absorbs part of the growth, so κ is a *biased* rate estimator. On rule 156's $D_{\max}$ it returns $b = 1.2554$ against a true $4^{1/5} = 1.31951$ — far enough off to look like a different constant, which is exactly how it first appeared here. Rates are therefore taken, in order of priority:

1. from an **exact integer recurrence** where one exists;
2. from an **analytic derivation** where R2 or R8 supplies one;
3. otherwise from the **two-parameter fit** $\ln y = c + \kappa N$, which cannot trade rate against power.

The two-parameter fit has the mirror-image bias, and the binomial rules expose it: with no α to absorb an $N^{-1/2}$ prefactor it pushes the base *below* 2 (1.9244 for W_{134}, whose $D_{\max}$ is exactly rule 150's central binomials). Neither fit is trustworthy alone, so the discriminating question is asked directly — is the residual $D_{\max}/2^N$ better explained by a power of N or by an exponential in N? When the power wins, the base is exactly 2 and α comes from that fit. That recovers $b = 2$ for the whole binomial family and puts them on the curve beside rule 150.

The comparison is made on the full series and not only on its tail, and it is refused when the series falls back down or when the fitted α is too steep to be a prefactor power ($|\alpha| > 8$). All three guards were added on 2026-07-31; the first changed eight of this report's $D_{\max}$ descriptors and is documented where they are listed (§8.2), the other two change nothing here and matter for R10.

5.5 Two bounds that the fitter needs and the theorem supplies

Since $1 \le n_{\mathrm{wcc}}, D_{\max} \le 2^N$, both bases lie in $[1, 2]$. A fit outside that window contradicts a theorem, so the theorem wins: 11 descriptors were clamped, each recorded, never silently. The clamp is exactly what the binomial rules need — W_{134}'s $D_{\max}$ is $924, 1716, 3003, 6435, 12870, 24310$, i.e. precisely rule 150's central binomials $\sim 2^N/\sqrt{N}$, and the $M2$ fit lands at 2.0588 because the $N^{-1/2}$ prefactor pulls the estimate past the boundary. Similarly a bounded integer series that has stopped changing is constant whatever the BIC says: W_{36}'s sector count is $9, 9, 10, 10, \ldots, 10$, which the unclamped fitter read as "exponential with base 0.95" — below 1, and impossible for a count.

The hyperbola as a predictor. Read backwards, (3) is sharper than any fit, and it called the answer here before the data could. At the $N \le 11$ stage the six rules 140, 156, 196, 198, 206, 220 all had an *exact* $a = \varphi$ with a $D_{\max}$ series still classified polynomial, so $ab = \varphi < 2$; the theorem already forced $b \ge 2/\varphi = 1.2361$. With the rate taken as §5.4 requires, all six now sit at

$$a\,b \;=\; \varphi \cdot 4^{1/5} \;=\; 1.61803 \times 1.31951 \;=\; 2.1350, \tag{5}$$

which is exactly the value V3 asks for — **above** the curve, with a from an exact Fibonacci recurrence and b from R2 §3's room-packing saddle point rather than from either fit. The six share one $D_{\max}$ series, $6, 9, 12, 16, 20, 27, 36, 48, 64, 81, 108$ over $N = 6 \ldots 16$, so the derivation covers all of them; the two candidate bases $3^{1/4} = 1.31607$ and $4^{1/5} = 1.31951$ differ by 0.26% and no finite series can separate them, which is precisely why the base is taken from theory.

6 Sectors against monitored attractors

6.1 The sector axis is uninformative for the V+reset rules

The left panel of Fig. 1 has a blunt message: 196 **of the** 254 **placeable rules sit at exactly** $(a, b) = (1, 2)$ — one sector, of size 2^N. Broken down by family:

family	rules	at $(1, 2)$	elsewhere
unitary (V only)	16	11	5
V + reset	160	**150**	10
V-free baseline	80	35	43
all	254	196 (77%)	58

So 94% of the rules that combine a Hadamard with a reset have *no sector structure at all*: the reset connects the whole basis into a single enclosure. That is a real result rather than a plotting problem, and it has a corollary for how this tier should be used. For those 150 rules the sector axis says nothing, and everything interesting is in the monitored map, where the same rules

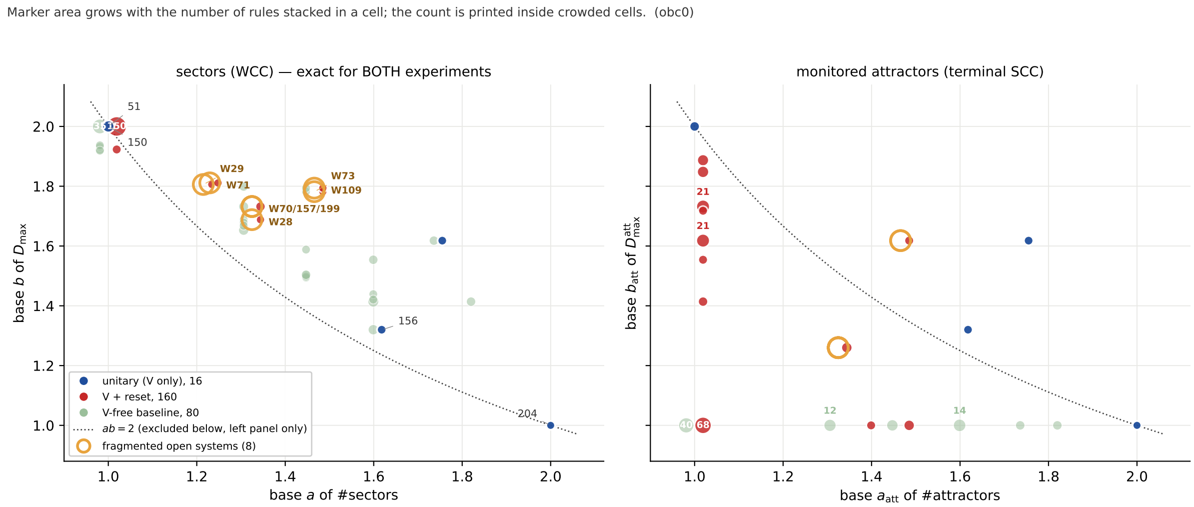

Figure 1: **F1, the two maps on identical axes.** Left: the sector (WCC) plane, base a of the sector count against base b of the largest sector — *exact for both experiments*. Right: the same plane for terminal SCCs — *monitored only*. The dotted line is $ab = 2$; it excludes the region below it in the left panel and is only a reference in the right one. Marker area grows with the number of rules stacked in a cell and the count is printed inside crowded cells, without which the distribution is invisible: 77% of the sector map lies in one cell. Colour is by family, with the quantum rules (those carrying a Hadamard) drawn forward and the V-free rules held back as a baseline. Gold rings and labels mark the eight V+reset rules that stay exponentially fragmented (§8.2). The nine unitary rules land at *identical* coordinates in both panels, which is assertion A2 made visible.

spread over the whole column $a_{\text{att}} \approx 1$ from $b_{\text{att}} = 1.0$ up to 1.89 (right panel of Fig. 1). The V-free baseline behaves the other way round: it is the family that populates the *sector* plane, and it collapses almost completely in the attractor plane.

Quantitatively, with the deficit $2 - a_{\text{att}} b_{\text{att}}$ of V4:

family	rules	median deficit
unitary (V only)	9	0.000
V + reset	152	+0.534
V-free baseline	78	+1.000

The unitary median is exactly zero because for a unitary rule the two planes *are* the same plane, and the median unitary rule sits on the curve. The V-free median of exactly +1 says $a_{\text{att}} b_{\text{att}} = 1$: their monitored attractors are $O(1)$ fixed points and essentially the whole space is transient. The V+reset family sits between the two, at +0.53.

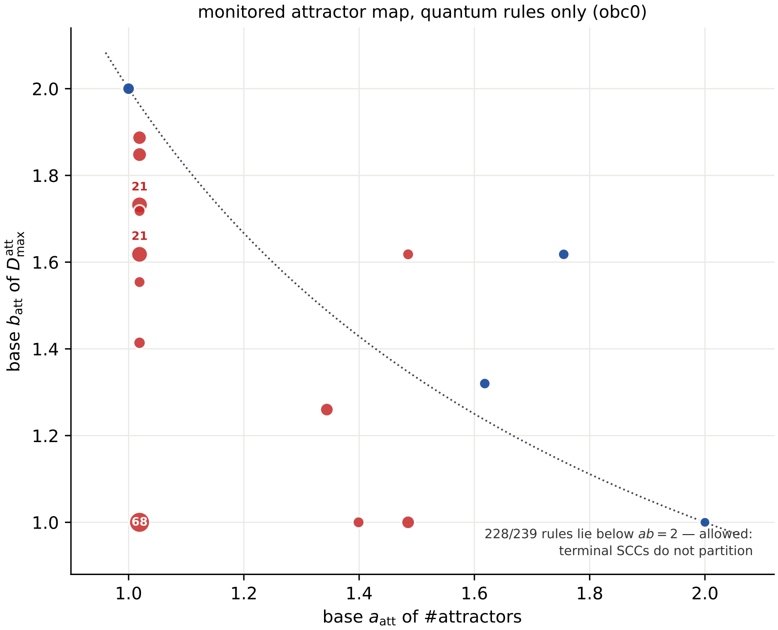

Figure 2: **F2, the monitored attractor map for the quantum rules alone** (unitary and V+reset; the V-free baseline is dropped here). MONITORED quantities throughout. This is where the V+reset family has structure: in the sector plane 150 of the 160 collapse onto $(1, 2)$.

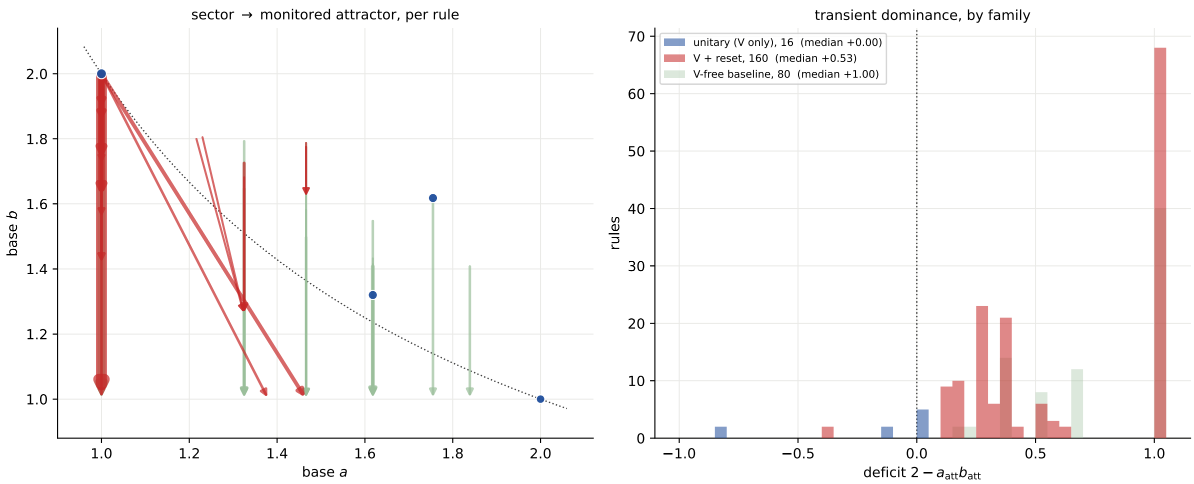

Figure 3: **F3, the difference between the two scatters, drawn as a difference.** Left: one arrow per rule, from its position in the sector plane to its position in the monitored-attractor plane; arrow width grows with the number of rules making the same move, and a dot means the rule does not move at all (which is what every unitary rule does, by A2). Right: the distribution of the deficit $2 - a_{\mathrm{att}}b_{\mathrm{att}}$ by family. The V-free baseline piles up at $+1$ — its attractors are $O(1)$ fixed points — while the unitary rules sit at 0.

rule	tuple	N	transient frac.	att./sector	$D^{\text{wcc}}_{\max}/2^N$
0	`DDDD`	16	1.0000	1.000	1.0000
2	`DVDD`	16	1.0000	1.000	1.0000
7	`EVDD`	16	1.0000	1.000	1.0000
8	`DIDD`	16	1.0000	1.000	1.0000
10	`DEDD`	16	1.0000	1.000	1.0000
15	`EEDD`	16	1.0000	1.000	1.0000
16	`DDVD`	16	1.0000	1.000	1.0000
18	`DVVD`	16	1.0000	1.000	1.0000
21	`EDVD`	16	1.0000	1.000	1.0000
24	`DIVD`	16	1.0000	1.000	1.0000
26	`DEVD`	16	1.0000	1.000	1.0000
32	`DDDV`	16	1.0000	1.000	1.0000
34	`DVDV`	16	1.0000	1.000	1.0000
37	`EDDV`	16	1.0000	1.000	1.0000
40	`DIDV`	16	1.0000	1.000	1.0000
42	`DEDV`	16	1.0000	1.000	1.0000

7 The corner view, and where the mass sits

Two things the base plane alone cannot show. The first is the sub-leading power: a rule with one giant sector and a rule with a *linear* number of them both sit at $a = 1$ (§8.5), and only $\alpha_{n_{\text{wcc}}}$ separates them. The second is the transient: the sector and attractor maps both plot counts and sizes, and neither says what fraction of the basis is still moving at late times.

7.1 Corner plots, as in R2 fig. 6

R2's growth map folds the base–base plane together with the two sub-leading powers by putting them in the margins. The main panel is base(count) against base($D_{\max}$); the bottom margin shares its x-axis and plots α_{count} beneath it; the left margin shares its y-axis and plots $\alpha_{D_{\max}}$ beside it. A rule therefore keeps its horizontal position between the main and bottom panels and its vertical position between the main and left panels, so one point carries the full (b, α) description of both series. The same layout is worth having for both Tier-1e maps.

7.2 Terminal SCCs against transient states

The recurrent set Rec is the union of the terminal SCCs; everything else is transient. Writing $|\text{Rec}| \sim c^N$, the base c answers the question directly: $c = 2$ means a finite fraction of the basis is still recurrent, and anything below means the transient part wins exponentially.
Three readings.

- **The unitary rules sit exactly at $c = 2$, alone.** Every state is recurrent, so $|\text{Rec}| = 2^N$ and the recurrent fraction is 1. That is a consistency check rather than a finding, but it fixes the top of the plot.
- **No dissipative rule is anywhere near it.** Not one of the 233 non-unitary rules with a descriptor has c within 0.02 of 2: adding a single reset makes the recurrent set an exponentially vanishing share of the basis. The median recurrent fraction at the largest N is 6.5×10^{-3} for the V+reset family and 2.6×10^{-4} for the V-free one.
- **The V-free family contracts somewhat harder**, which is the opposite of what one might guess from the fact that it has no quantum gate, but the effect is modest: 51% of V-free rules have a *bounded* recurrent set ($c = 1$, a fixed handful of frozen states at any

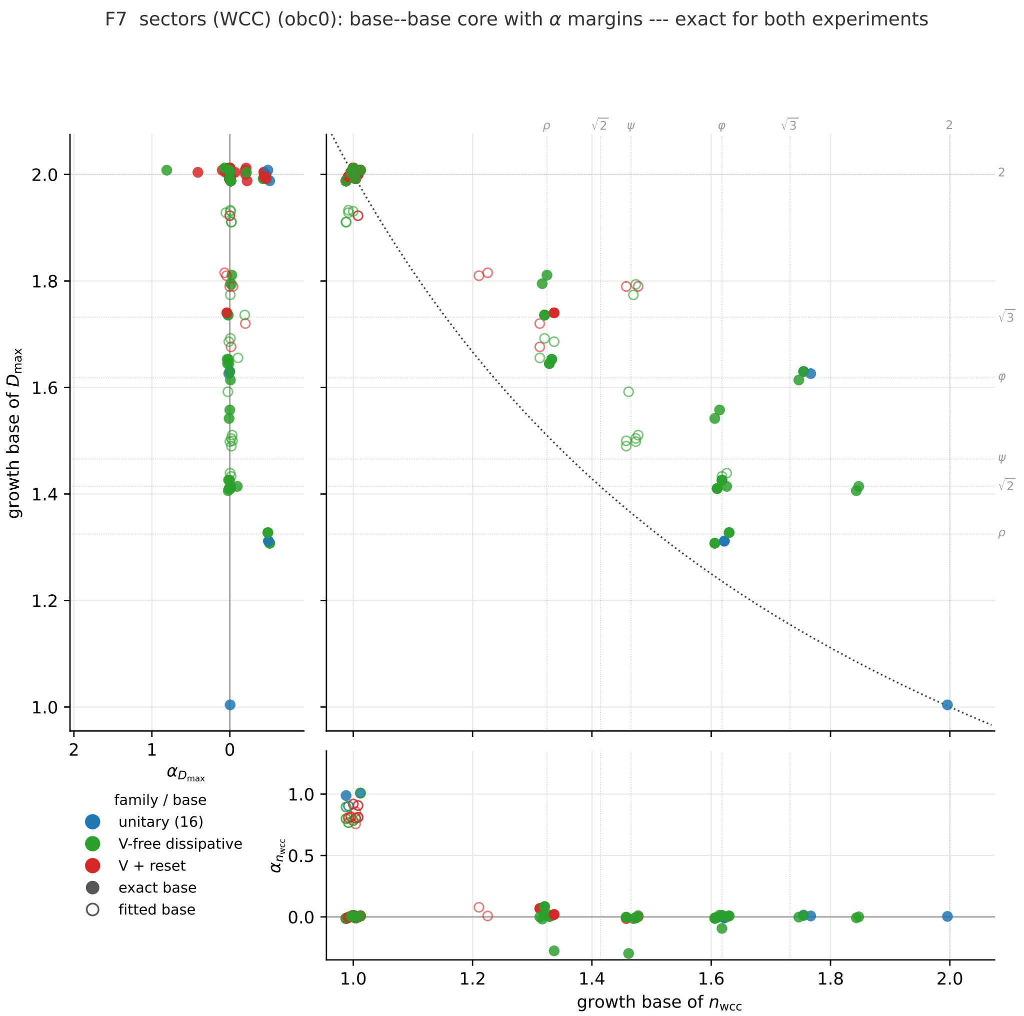

Figure 4: **F8, the sector map in corner form** (`obc0`). Main panel: the map of Figure 1 with the exclusion curve. Bottom margin: the sector count's sub-leading power against its base — the column at $a = 1$ splits into $\alpha \simeq 0$ (genuinely one sector) and $\alpha \simeq 1$ (the pinned-frontier rules of §8.5, a linear number of them), which the main panel cannot distinguish. Left margin: the $D_{\max}$ power, where the binomial family's $-\frac{1}{2}$ is visible at $b = 2$. Filled marker = exact base.

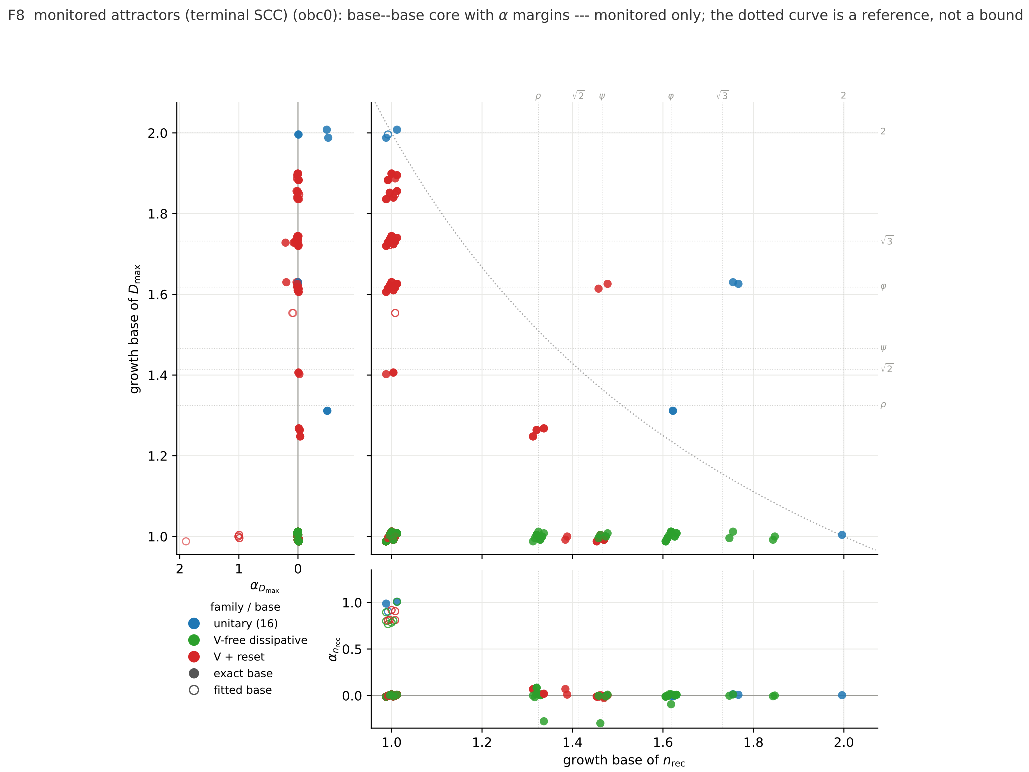

Figure 5: **F9, the same for the monitored attractors** (`obc0`), i.e. terminal SCCs. *Monitored only.* The dotted curve is drawn as a reference and is *not* a bound here — terminal SCCs do not partition the basis — and most rules do lie below it. The V-free family (green) collapses onto $b_{D_{\max}} = 1$: their attractors are single states or short cycles however many there are.

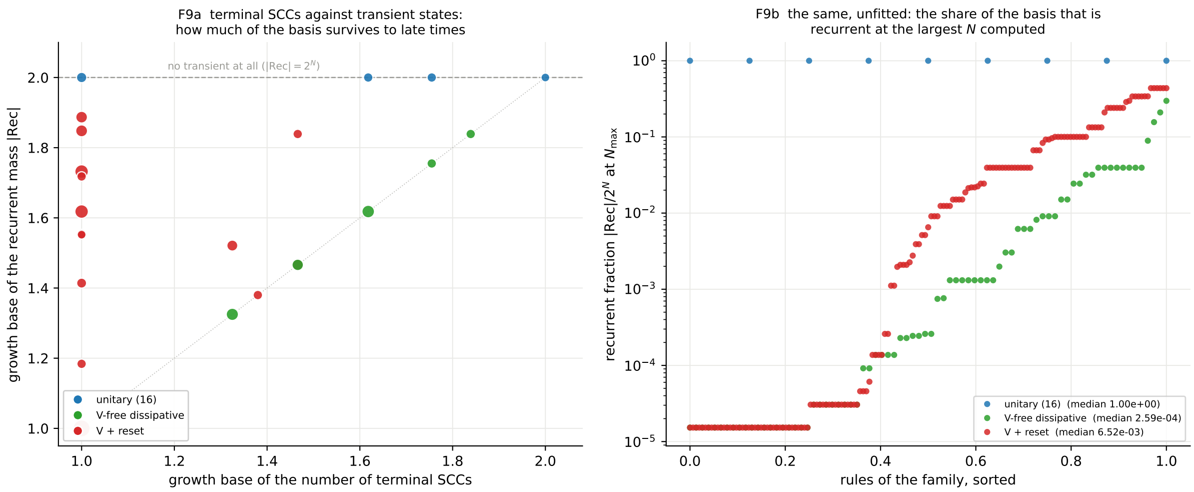

Figure 6: **F10, where the mass sits** (`obc0`, monitored). Left: growth base of the number of terminal SCCs against the growth base of the recurrent *mass*. Right: the same without any fit — the recurrent fraction at the largest N computed, each family sorted.

N) against 44% of the V+reset rules, and the two base distributions separate at $p = 0.006$ on a rank test ($p = 0.013$ on the log recurrent fraction). The Hadamard, by spreading a state over its sector, keeps more of the basis alive — but by about a decade in the median fraction, not by an exponent.

- **And most of that gap is one subfamily.** Splitting the V-free rules at whether the rule is all-reset:

V-free subfamily	rules	bounded	median c	median $\lvert\mathrm{Rec}\rvert/2^N$
all-reset (the 16 boolean $f(\ell, r)$)	14	12	1	1.5×10^{-5}
reset + identity	64	28	$\rho = 1.3247$	1.3×10^{-3}

  1.5×10^{-5} is 2^{-16}: a *single* recurrent state. An all-reset rule keeps no memory of the previous configuration, so it collapses to a fixed point. Remove those 14 and the rest of the V-free family sits within a factor of 5 of the V+reset one. The extreme contraction is a property of memorylessness, not of V-freeness.

A warning about medians on this plot. An earlier draft of the third bullet quoted the *median mass base* of each family: exactly 1 for V-free against $\psi = 1.46557$ for V+reset. Both numbers are correct and both are worthless. The fitted bases are not a continuous spread — every one of them is an algebraic number, and they pile up on a few atoms (68 V+reset rules at 1, then 21 at φ, 21 at $\sqrt{3}$, 9 at 1.887, 8 at 1.848, and only 6 at ψ). The median is a rank statistic, and rank 78 of 155 simply falls inside the six-rule ψ block; landing on a named constant carries no information when every candidate value is one. It is also unstable exactly where it was being used: the 68 rules at base 1 are ten short of a majority, so moving ten rules would put the V+reset median at 1 as well and erase the contrast, while the V-free median of 1 rests on 40 of 78 — a majority of one rule. Quantities with atoms of this size want a proportion or a rank test, which is what the bullet now reports; the recurrent *fraction* is continuous and needs no such care.
A caution on what F9 and F10 are. Both are statements about the *monitored* circuit: the recurrent/transient split is a property of the measured chain, and the unmonitored channel can keep weight on states the graph calls transient (R5's rule 22, R7's immortal transient). The sector map F8 is the one that is exact for both experiments.

7.3 Correction (2026-08-01): α and the base came from different models

Drawing these margins next to R2's exposed a defect in the fitting layer that the base-only maps had hidden for the whole project. The descriptor reports a *pair*, $y \sim c\,N^{\alpha} b^{N}$, and the two halves were being taken from different models. BIC fits M2, $\ln y = c + \alpha \ln N + \kappa N$, whose α is whatever compensates *its own* κ. Every branch that runs afterwards then replaces the base — with an exact integer recurrence, a parity recurrence, the pure-exponential refit of §5.4, or a clamp — and the inherited α was left describing a base that was no longer there.

Rule 28's $D_{\max}$ is the cleanest case. The series is

$$4,\ 4,\ 8,\ 8,\ 8,\ 16,\ 16,\ 16,\ 32,\ 32,\ 32 \qquad (N = 6 \ldots 16),$$

exactly $c \cdot 2^{N/3}$ — a pure staircase with *no* prefactor. The recurrence correctly pins the base at $2^{1/3} = 1.2599$, and the descriptor then reported $\alpha = 2.34$ alongside it, a spurious $N^{2.34}$ produced entirely by M2's abandoned κ. Rules 39 and 47 carried a fictitious $\alpha = 1.24$ the same way.

The fix is to hold the base that is actually being reported fixed and regress the residual on $\ln N$ (`sectors._alpha_at_base`). Rule 28 then gives $\alpha = -0.03$, rule 39 gives 0.08, rule 47 gives

0.03: correctly flat, as the series plainly are. In the sector map 111 of 512 α values move, by a median of 0.10 and a maximum of 1.99.

Nothing else moves. No base changes, not one, and no verdict of §5 changes; the defect was confined to the sub-leading power, which is why the base-only maps of R2 and §6 were never wrong. Both readings this section makes survive unchanged: the $a = 1$ column still splits into 173 rules at $\alpha \simeq 0$ and 30 at $\alpha > 0.5$, containing all eight pinned-frontier rules, and the binomial family still carries $\alpha = -\frac{1}{2}$ at $b = 2$ (analytic for 150/105, and -0.43 from the volume-fraction discriminator for 134).

R2's growth map had the same defect, and worked around it. Its `growth_map._dissipative_descript` set $\alpha = 0$ by fiat for every rule it classed exponential, on the stated grounds that "the sub-leading power is not identifiable under a period split". The diagnosis was too pessimistic — the power is perfectly identifiable once the base is pinned; what was unidentifiable was the *stale* α — but the workaround was the right call given the value on offer, and it is why R2's fig. 6 margins are a column of structural zeros for 440 of 486 series. With both sides repaired, the two figures agree: the 63 disagreements above 0.15 fall to 2, and those two are rules 105 and 150, where R9 uses R8's derived linear sector count ($\alpha = 1$ exactly) and R2 still fits 0.83. Derivation beats fit, so R9's value stands.

A related defect fixed at the same time: `sector_figure`'s `__main__` guard sat in the middle of the module, before `fig_corner` and `fig_recurrent_transient` were defined, so `python -m ...sector_figure` raised `NameError` after writing F1–F6 and left F8–F10 stale on disk. Both are now pinned by tests.

8 Clustering the dissipative rules by their coherent correspondent

Every V+reset rule has a *coherent part*. A symbol-table entry says what happens at a site for one neighbour pattern: `I` does nothing, `V` applies the Hadamard, `D` resets to $|0\rangle$, `E` resets to $|1\rangle$. Switching the dissipation off means the reset entries stop acting,

$$\texttt{D},\ \texttt{E} \longmapsto \texttt{I}, \tag{6}$$

which leaves only `I` and `V` — always one of the sixteen unitary rules. So (6) assigns each dissipative rule a unitary *correspondent*, and the assignment partitions the rule space exactly: a unitary parent with v Hadamard entries has

$$3^{4-v} - 1 \tag{7}$$

V+reset children, each non-`V` entry independently becoming `I`, `D` or `E`, minus the all-`I` case which is the parent itself. Summing over the fifteen unitary rules with $v \geq 1$ gives $4{\cdot}26+6{\cdot}8+4{\cdot}2+0 = 160$, the whole V+reset family, in 14 non-empty clusters (`VVVV` has no `I` to spoil, so it has no children). This is verified against the rule tables rather than assumed.

8.1 What adding dissipation does

parent	tuple	a	b	children	$\to (1,2)$	at parent	keeps structure
156	`IIVI`	1.6180	1.3195	26	22	0	**0**
198	`IVII`	1.6180	1.3195	26	22	0	**0**
108	`IIIV`	1.7549	1.6180	26	25	0	**0**
201	`VIII`	1.7549	1.6180	26	25	0	**0**
54	`IVVV`	1.0000	2.0000	2	2	2	—
57	`VIVV`	1.0000	2.0000	2	2	2	—
60	`IIVV`	1.0000	2.0000	8	8	8	—
99	`VVIV`	1.0000	2.0000	2	2	2	—
102	`IVIV`	1.0000	2.0000	8	8	8	—
105	`VIIV`	1.0000	2.0000	8	8	8	—
147	`VVVI`	1.0000	2.0000	2	2	2	—
150	`IVVI`	1.0000	2.0000	8	8	8	—
153	`VIVI`	1.0000	2.0000	8	8	8	—
195	`VVII`	1.0000	2.0000	8	8	8	—
all				160	150	56	

The table splits the fourteen clusters cleanly in two.

- **Ten parents already sit at** $(1,2)$ — one sector holding the whole space — so there is nothing for a reset to destroy, and all 56 of their children stay exactly where the parent is.
- **Four parents have genuine sector structure** — the $v = 1$ rules 156, 198, 108, 201 — and they hold 104 children between them. *Not one of the* 104 *keeps its parent's position.* 94 collapse all the way to $(1,2)$; the remaining 10 move to some intermediate point but never stay — and eight of those ten are still *exponentially* fragmented, which is §8.2.

That is the sharpest statement this tier produces about dissipation: **adding any reset to a rule that has sector structure destroys that structure, without exception over the** 104 **cases available**. It also explains the pile-up of §6.1 arithmetically — the 150 V+reset rules at $(1,2)$ are exactly the 94 that collapsed plus the 56 that were never anywhere else.

8.2 Fragmentation in an open quantum system

The 104 children of the four structured parents split 94 / 10: ninety-four collapse to $(1,2)$, and ten land somewhere in between. Those ten are the interesting ones, and eight of them are interesting for a specific reason — their sector count still grows *exponentially*.

rule	tuple	parent		a	name	exact	b	name	ab
73	`VIID`	201	`VIII`	1.46557	supergolden ψ	✓	1.78187	—	2.6115
109	`EIIV`	108	`IIIV`	1.46557	supergolden ψ	✓	1.79363	—	2.6287
28	`IIVD`	156	`IIVI`	1.32472	plastic ρ	✓	1.68818	—	2.2364
70	`IVID`	198	`IVII`	1.32472	plastic ρ	✓	1.73205	$\sqrt{3}$	2.2945
157	`EIVI`	156	`IIVI`	1.32472	plastic ρ	✓	1.73205	$\sqrt{3}$	2.2945
199	`EVII`	198	`IVII`	1.32472	plastic ρ	✓	1.73205	$\sqrt{3}$	2.2945
71	`EVID`	198	`IVII`	1.22937	—	fit	1.81132	—	2.2268
29	`EIVD`	156	`IIVI`	1.21470	—	fit	1.80585	—	2.1936

Why this is worth stating as a result rather than a curiosity. A sector is an *enclosure*: span(WCC) is invariant under every Kraus operator, so an exponentially growing sector count is a statement about the unmonitored channel just as much as about the measured circuit (§1). These eight rules therefore exhibit **Hilbert-space fragmentation as open quantum systems** — not a unitary circuit that someone later added noise to, and not an artefact of conditioning on measurement records. Each contains a genuine reset, and each has a coherent correspondent that is itself fragmented, so the fragmentation is inherited rather than accidental.

The structure of the list is tidier than the selection deserves:

- They come from all **four** of the structured parents — the $v = 1$ rules 156, 198, 108, 201 — three, three, one and one respectively. No $v \geq 2$ parent contributes, because none of them has sector structure to pass on in the first place.
- The bases are **exact algebraic numbers**, from integer recurrences, for six of the eight: the **plastic number** $\rho = 1.32472$, the real root of $x^3 = x + 1$, for 28, 70, 157, 199; and the **supergolden ratio** $\psi = 1.46557$, the real root of $x^3 = x^2 + 1$, for 73 and 109. For the four ρ rules the reset *raises the degree* of the characteristic polynomial: their parents 156 and 198 count sectors by $x^2 = x + 1$ and the children count by $x^3 = x + 1$. That pattern does *not* extend to 73 and 109, whose parents 201 and 108 already count by a cubic, $x^3 = 2x^2 - x + 1$; there the degree is unchanged and only the polynomial moves, to $x^3 = x^2 + 1$. So there is no single statement covering all eight — see §8.3.
- They pair up. $\{29, 71\}$ share $\approx (1.22, 1.81)$, and 73, 109 are each their own reflection partner. Within a pair the two rules are reflections of one another, one built on `D` and the other on `E`. On the b axis the four ρ rules collapse onto a *single* value: 70 and 199 — and their reflections 78 and 157 — all have $D_{\max} = 2 \cdot 3^{N/2-1}$ on even N, i.e. $b = \sqrt{3}$ exactly.
- Rule 28 (and its reflection 92) is the one case in the list where the two parities do not share a base. Its $D_{\max}$ is $14, 28, 38, 76, 108, 208, 324, 568, 972, 1552, 2916$; the even branch settles to a ratio of exactly 3 per two sites, so $b_{\text{even}} = \sqrt{3}$ like the others, while the odd branch settles to $1 + \sqrt{3}$ per two sites, i.e. $b_{\text{odd}} = \sqrt{1 + \sqrt{3}} = 1.6529$. The single number in the table, 1.68818, is a fit through both branches and should be read as "between the two". Rule 28 is also the one whose φ prediction §4 falsified — and the plastic number that replaced it is the same constant that appears here for three other rules.

Correction, 2026-07-31. An earlier version of this section reported $b = 2$ *exactly* for 28 and 70, and $b = 2$ for 132, 133, 164, 222, 78 and 92 as well: eight $D_{\max}$ descriptors in all. That came from the volume-fraction discriminator (§5.4), which decides whether $D_{\max} = c\,2^N N^\alpha$ by asking whether $D_{\max}/2^N$ is better explained by a power of N than by an exponential, and which was being run on a six-point tail. Over so short a window $\ln N$ is nearly affine in N and the power-law model can win against an exponential decay that is genuinely there. Requiring it to win on the *full* series as well removes all eight, and the corrected numbers are the ones tabulated above. The direction of the error was uniform: each of the eight has a $D_{\max}$ that is an exponentially shrinking fraction of the space — W_{222} falls from 0.328 of the basis at $N = 6$ to 0.010 at $N = 16$ — and none was a fixed fraction. No verdict changes: all eight were already *above* the exclusion curve and remain so, with products now 2.20–2.33 instead of 2.65–3.24. The guard was found while applying the same fitting layer to the X-gate cycle series in R10 §3, where its failure mode is far more visible.

The remaining two of the ten, 6 (`IVDD`) and 20 (`IDVD`), have $a = 1$: their sector counts are linear $(4, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9)$, so they keep a little structure but are not fragmented.

A caution on the two fitted rows. For 29 and 71 the base comes from a parity-resolved fit, not a recurrence, so 1.2147 and 1.2294 should be read as "about 1.22" and not as two distinct constants; their series oscillate strongly with N $(3, 6, 5, 9, 7, 14, 10, 21, 15, 31, 22$ for rule

29) and each parity branch has only five or six points. That they are exponential is solid — both branches grow geometrically — but the value is not pinned, and if they follow the pattern of the other six the honest guess is that both are the same algebraic number.

8.3 Which constant describes what: a naming correction

Earlier drafts of this report called the four structured parents the "φ-fragmented" rules. That is wrong, and worth correcting explicitly because the error is easy to make: φ describes *different observables* in the two pairs, and no single constant describes a rule's fragmentation at all. The exact recurrences:

rule	sector count n_{wcc}	largest sector $D_{\max}$
156, 198	$21, 34, 55, 89, \ldots$	$6, 9, 12, 16, 20, 27, \ldots$
	Fibonacci, $a_n = a_{n-1} + a_{n-2}$	*no* integer recurrence
	base φ, root of $x^2 = x + 1$	base $4^{1/5}$, from theory (R2 §3)
108, 201	$37, 65, 114, 200, \ldots$	$8, 13, 21, 34, 55, \ldots$
	$a_n = 2a_{n-1} - a_{n-2} + a_{n-3}$	Fibonacci, $a_n = a_{n-1} + a_{n-2}$
	base 1.75488, root of $x^3 = 2x^2 - x + 1$	base φ

Two separate problems with the old label.

- **Across rules, φ changes role.** For 156 and 198 it is the base of the sector *count* — there are Fibonacci-many sectors. For 108 and 201 it is the base of the largest sector *size* — the biggest sector is Fibonacci-sized — while their sector count grows as the root of $x^3 = 2x^2 - x + 1$. Calling both pairs "φ-fragmented" asserts a shared property they do not have.
- **Within one rule, the two quantities have unrelated mechanisms.** Rule 156's sector count is a Fibonacci word count; its largest sector is *not* an integer recurrence at all, and its base comes from the room-packing saddle point of R2 §3, an optimisation over room length that has nothing to do with φ. Naming the rule after one constant hides the fact that its two structural laws are different laws.

So the group is referred to below as **the four exponentially fragmented parents**, and any base is quoted together with the observable it belongs to. The pairs are, in the notation (a, b) of §5,

$$156, 198: \ (\varphi,\ 4^{1/5}),\ ab = 2.1350; \qquad 108, 201: \ (1.75488,\ \varphi),\ ab = 2.8394,$$

so both sit above the exclusion curve, but for structurally different reasons — the first pair by having many sectors, the second by having large ones. That φ appears in both lines is a coincidence of which observable one looks at, not evidence of a common mechanism.

8.4 The base plane hides a whole tier, and 150/105 are in it

The bookkeeping above is done in the base plane, and the base plane cannot tell a linear sector count from a single sector: both give $a = 1$. So rule 150 — which has $\lfloor (N+1)/2 \rfloor + 1$ sectors, nine of them at $N = 16$, the domain-wall charge of R8 — is plotted on top of rule 51, which has exactly one. Calling both of them "already at $(1, 2)$, nothing to destroy" is wrong for 150. Survival has to be asked in terms of the growth *class*.

parent's sector growth	parents	#	children	→exp	→poly	→const
exponential	108, 156, 198, 201	4	104	8	14	82
polynomial	60, 102, 105, 150	4	32	0	0	32
constant	54, 57, 99, 147, 153, 195	6	24	0	0	24

Three quite different fates, which the $(1, 2)$ cell had conflated:

- **Exponential parents** (108, 156, 198, 201), 104 children: 8 stay exponential (§8.2), 14 **degrade to polynomial but keep growing**, 82 collapse to a constant number of sectors.
- **Polynomial parents** — 150 and 105 with the wall charge ($\alpha = 1$), 60 and 102 with the frontier charge of §8.5 (fitted $\alpha \approx 0.82$, same dyadic tower) — 32 children: 0 **survive.** Every one drops to a *constant* sector count. For 150 specifically, six of its eight children fall to a single sector and the other two (22 `IVVD`, 182 `IVVE`) to two sectors; all eight of 105's fall to one. The domain-wall charge does not tolerate a reset anywhere.
- **Constant parents** (six rules), 24 children: nothing to lose — though note even rule 54's two sectors become one.

So the answer for 150 and 105 is blunt: **nothing survives**. That is a *stronger* statement than for the exponential parents, where $8 + 14$ of 104 children keep some growth. Polynomial sector structure is more fragile than exponential sector structure, not less.

But the wall charge is a sink, not a source. The fourteen polynomial survivors are worth looking at, because six of them have *exactly* rule 150's sector series $4, 5, 5, 6, 6, 7, 7, 8, 8, 9, 9$ and two have rule 105's:

rule	tuple	parent	n_{wcc}, $N{=}6 \to 16$	series
44	`IIDV`	108	$7 \to 17$	—
100	`IDIV`	108	$9 \to 19$	—
104	`DIIV`	108	$4 \to 9$	$= 105$
110	`IEIV`	108	$7 \to 17$	—
124	`IIEV`	108	$7 \to 17$	—
20	`IDVD`	156	$4 \to 9$	$= 150$
148	`IDVI`	156	$4 \to 9$	$= 150$
158	`IEVI`	156	$4 \to 9$	$= 150$
188	`IIVE`	156	$7 \to 17$	—
6	`IVDD`	198	$4 \to 9$	$= 150$
134	`IVDI`	198	$4 \to 9$	$= 150$
214	`IVEI`	198	$4 \to 9$	$= 150$
230	`IVIE`	198	$7 \to 17$	—
233	`VIIE`	201	$4 \to 9$	$= 105$

Those six are 20, 148, 158 (from 156) and 6, 134, 214 (from 198) — and four of them are exactly the mixed rules of §9 that share 150's whole multiset. So dissipation can degrade the exponential fragmentation of 156/198 *down to* the rule-150 wall charge, while a reset applied *to* the wall charge destroys it outright. The wall charge sits one level below the exponential rules in a hierarchy

$$\text{exponential count } (\varphi \text{ or } 1.75488) \longrightarrow \underbrace{\rho,\ \psi}_{8} \longrightarrow \underbrace{\text{wall charge}}_{14} \longrightarrow \underbrace{\text{one sector}}_{82},$$

and it is a terminal rung: reachable from above, but with no dissipative descendants of its own.

8.5 A second polynomial mechanism: the pinned frontier

Six of the fourteen polynomial survivors — 44, 100, 110, 124, 188, 230 — are *not* degraded copies of the rule-150 wall charge. They carry a different conserved quantity, and two *unitary* rules, $60 = \texttt{IIVV}$ and $102 = \texttt{IVIV}$, carry the same one, so it is not produced by dissipation at all. The sector sizes give it away immediately. At $N = 12$, rule 110 has thirteen sectors of sizes

$$2048,\ 1024,\ 512,\ 256,\ 128,\ 64,\ 32,\ 16,\ 8,\ 4,\ 2,\ 1,\ 1,$$

a **dyadic tower** rather than a binomial row. The charge is the **position of the extremal excitation**: verified at $N = 9$, every sector is a level set of the index of the rightmost 1 for 110, 230 and 102, and of the leftmost 1 for 124, 188, 44 and 60. The outermost excitation is pinned by the reset and everything inside it is free, which gives

$$n_{\text{wcc}} = N + 1, \qquad D_{\max} = 2^{N-1} \text{ exactly}, \qquad \frac{D_{\max}}{2^N} = \tfrac{1}{2}. \tag{8}$$

rule	tuple	parent	pinned	n_{wcc}	at N=16	$D_{\max}$	$D_{\max}/2^N$
110	`IEIV`	108	rightmost	N+1	17	2^{N-1}	0.5000
124	`IIEV`	108	leftmost	N+1	17	2^{N-1}	0.5000
188	`IIVE`	156	leftmost	N+1	17	2^{N-1}	0.5000
230	`IVIE`	198	rightmost	N+1	17	2^{N-1}	0.5000
44	`IIDV`	108	leftmost (+split)	parity	17	2^{N-1}	0.5000
100	`IDIV`	108	leftmost (+tail)	N+3	19	—	0.5000
60	`IIVV`	60	leftmost (unitary)	N+1	17	2^{N-1}	0.5000
102	`IVIV`	102	rightmost (unitary)	N+1	17	2^{N-1}	0.5000

Set against the wall charge, the contrast is sharp, and both are invisible in the base plane because both have $a = 1$, $b = 2$:

	rule-150 family	frontier family
conserved quantity	domain-wall number	position of the extremal excitation
n_{wcc}	$\lfloor (N+1)/2 \rfloor + 1 \sim N/2$	$N + 1$
sector sizes	binomial $\binom{N+1}{w}$, even w	dyadic $2^{N-1}, \dots, 2, 1, 1$
$D_{\max}$	$\binom{N+1}{\sim N/2} \sim 2^N/\sqrt{N}$	2^{N-1} exactly
$D_{\max}/2^N$	$\sim N^{-1/2} \to 0$	$\frac{1}{2}$, constant

The frontier family has *twice* as many sectors asymptotically and a largest sector that stays a fixed half of the space rather than shrinking. Two of the six dissipative members are less clean: 44 splits each frontier class further at odd N ($n_{\text{wcc}} = N+1$ for even N, $N+3$ for odd), and 100 has $N+3$ sectors with a short non-dyadic tail $(\dots, 26, 16, 6, 6, 4, 2)$, so its charge is the leftmost excitation plus a refinement.

Two of them are unitary, which reclassifies them. 60 and 102 were listed in §8.4 among the polynomial parents beside 150 and 105, on the strength of a fitted $\alpha \approx 0.82$. That grouping is wrong: their sector sizes are the same dyadic tower and their charge is the same extremal excitation, so they belong here, and the differing α only reflects the fitter seeing one series through a different window. It also means the frontier charge exists already in the *unitary* rule space — it is a destination reachable both directly (60, 102) and as a dissipative degradation of the exponential rules (110, 124, 188, 230 from 108, 156, 198) — and not something a reset creates.

Since sectors are enclosures, this is an open-system statement in the sense that matters here: the pinned frontier is a conserved quantity of the *channel*, not of a measurement record. It is a genuinely different route to polynomial fragmentation than the wall charge, and both are reachable from the same exponentially fragmented parents — 108 contributes four frontier rules while 156 and 198 contribute the wall-charge ones.

A caveat this report cannot settle. The charge here is the position of the *extremal* excitation, and that is a quantity only an open chain has — a ring has no leftmost site. So the frontier structure is, on its face, tied to the boundary in a way the domain-wall number of §9 is not: the wall number is defined by $b_j = x_j \oplus x_{j+1}$ everywhere in the bulk, whereas "the first 1 from the left" presupposes a left. Whether the frontier sectors are bulk structure or an artefact of `obc0` is therefore a real question, and it is not one this report can answer, because answering it means changing the boundary condition. It is deferred to the `pbc` report, and until then §8.5 should be read as a statement about open chains only. The same question does not arise for the eight exponentially fragmented rules of §8.2, whose charges are not defined by reference to an end, but confirming that is also `pbc` work.

And 105 is not 150's partner. R8 treats them as reflection partners and this report inherited that, but the sector multisets differ at $N \equiv 1 \pmod 4$. Rule 105 = `VIIV` fires when the two neighbours *agree*, where 150 = `IVVI` fires when they differ, and at those N the sectors of 105 carry *odd* wall number: at $N = 9$ its multiset is $[252, 120, 120, 10, 10] = \binom{10}{\text{odd } w}$ against 150's $[210, 210, 45, 45, 1, 1] = \binom{10}{\text{even } w}$, and likewise at $N = 13$. At all other N in range the two agree exactly. The growth is linear either way, which is all the base captures — another thing the base plane hides.

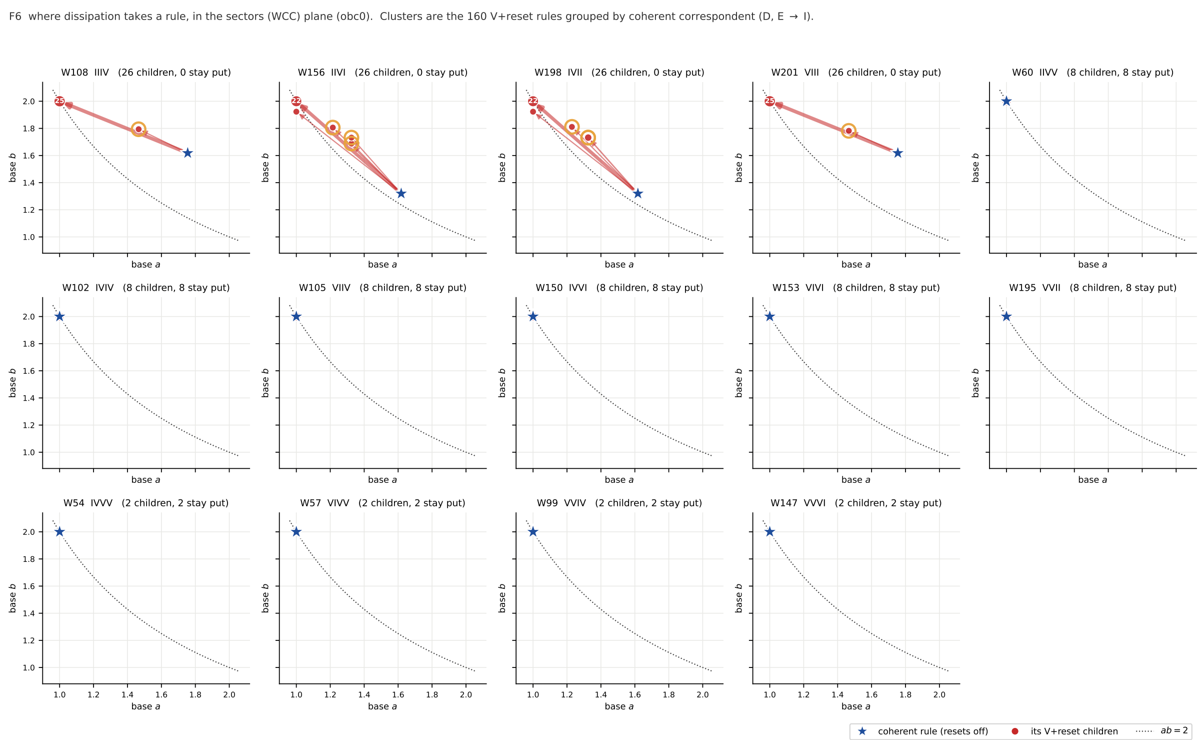

Figure 7: **F6, the move in the SECTOR plane, clustered by coherent correspondent.** One panel per unitary rule: the star is the coherent rule (all resets switched off, (6)), the dots are its V+reset children, and each arrow is the move that turning the resets on produces. Dot area and arrow width grow with the number of children making the same move. Top row and the two leftmost middle panels: parents with sector structure, every arrow pointing away and up-left towards $(1, 2)$. Remaining panels: parents already at $(1, 2)$, where the children coincide with the parent and no arrow is drawn. Gold rings mark the eight children that remain exponentially fragmented (§8.2). These coordinates are exact for both experiments.

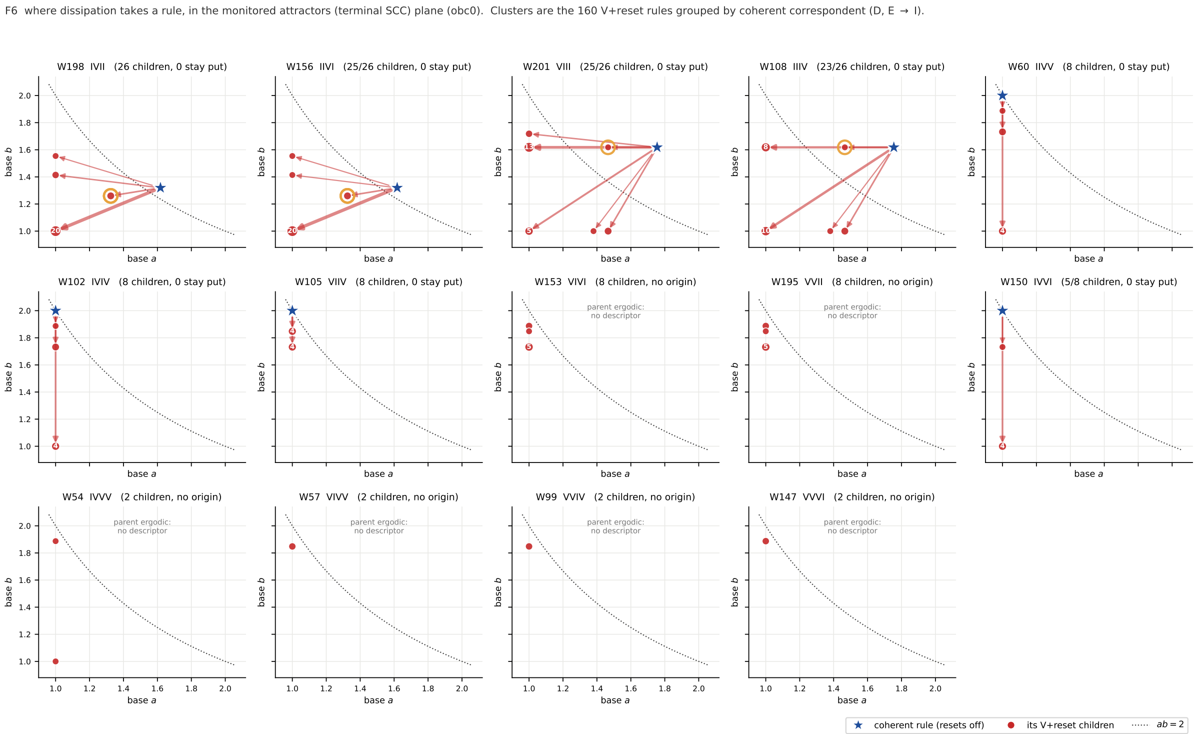

Figure 8: **F7, the same clusters in the MONITORED attractor plane.** Here the moves are large for every cluster, including the eight whose sector coordinates do not move at all — which is the point of keeping both planes: dissipation is invisible on the sector axis for those rules and plainly visible on the attractor axis. Six panels have no star: those parents (54, 57, 99, 147, 153, 195) are ergodic at `obc0`, so Tier 1a never produced an attractor descriptor for them and the cluster has no origin to draw from. Monitored quantities throughout.

9 Rule 150's wall charge survives six dissipative rules

The sector axis makes a comparison possible that the attractor axis could not, and it turns up an exact coincidence. Consider the nine rules of the form `I··I`, i.e. those where a site is inert unless its two neighbours differ. Seven of them — including six *dissipative* ones — have **exactly** rule 150's sector-size multiset, at every $N = 6 \ldots 16$:

rule	tuple	sector multiset = that of W_{150}
134	`IVDI`	✓
142	`IEDI`	✓
148	`IDVI`	✓
150	`IVVI`	— (unitary reference)
158	`IEVI`	✓
212	`IDEI`	✓
214	`IVEI`	✓
132	`IDDI`	$\times$ ($n_{\text{wcc}} = 2584$, $D_{\max} = 322$ at $N = 16$)
222	`IEEI`	$\times$ ($n_{\text{wcc}} = 988$, $D_{\max} = 686$ at $N = 16$)

So the domain-wall charge of R8 — $|S_w| = \binom{N+1}{w}$ on even w, $\lfloor (N+1)/2 \rfloor + 1$ sectors — is not a property of unitarity at all. It survives replacing *either or both* Hadamards by a reset, and the two exceptions are precisely the rules whose two active symbols are the *same* reset type (`DD` or `EE`). This is the sector-level counterpart of the Tier-2 observation (R-T13) that W_{148} and W_{134} inherit an exact decoherence-free subspace of dimension $\lfloor (N+1)/2 \rfloor + 1$, which is exactly C150's sector count: here the whole multiset, not just its cardinality, is shown to agree, and for six rules rather than two.

A conjecture that fails. The obvious generalisation — that the sector partition depends only on which of the four symbols are `I`, since `V`, `D` and `E` all generate the same *undirected* flip edge $x \leftrightarrow x \oplus 2^i$ — is **false**. Grouping all 256 rules by their `I`-pattern gives 16 groups, and only three collapse to a single sector signature: $\{140, 156, 220\}$, $\{196, 198, 206\}$ and the singleton $\{204\}$. (Those first two are exactly the six rules of §5.5 whose *sector count* has base φ.) The reason the argument fails is the brick-wall composition: the odd-layer symbol is read off the *post-even-layer* state, and a `V` leaves a branch supported on both values of the flipped bit where a `D` or `E` leaves only one, so the downstream firing pattern differs even though the single-site undirected edge does not.

10 Growth classes and exact bases

series	family	const	poly	exp	irreg	exact base
#sectors	unitary	7	4	5	0	14
#sectors	classical	28	12	38	2	66
#sectors	mixed	138	14	8	0	143
$D_{\max}$	unitary	1	0	15	0	16
$D_{\max}$	classical	0	0	78	2	59
$D_{\max}$	mixed	0	0	160	0	152

rule	tuple	a	name	source	b	ab
204	`IIII`	2.00000	2	IIII: every state frozen	1.0000	2.0000
76	`IIID`	1.83929	tribonacci ψ	integer recurrence	1.4142	2.6011
205	`EIII`	1.83929	tribonacci ψ	integer recurrence	1.4142	2.6011
108	`IIIV`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
200	`DIII`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
201	`VIII`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
236	`IIIE`	1.75488	root of $x^3 = 2x^2 - x + 1$	integer recurrence	1.6180	2.8395
232	`DIIE`	1.61803	golden φ	integer recurrence	1.5538	2.5141
4	`IDDD`	1.61803	golden φ	integer recurrence	1.4142	2.2882
12	`IIDD`	1.61803	golden φ	integer recurrence	1.4142	2.2882
68	`IDID`	1.61803	golden φ	integer recurrence	1.4142	2.2882
132	`IDDI`	1.61803	golden φ	integer recurrence	1.4391	2.3285
218	`DEEI`	1.46557	supergolden ψ	integer recurrence	1.5879	2.3272
28	`IIVD`	1.32472	plastic ρ	integer recurrence	1.6882	2.2364

11 Basins, and a gap in the Tier-1a archive

Basins are a **monitored** observable throughout this section.

The sum rule $\sum_k |\text{basin}_k| + |\text{shared}| = 2^N$ turned out to be *unanswerable* for 76 `obc0` units. The cause is a schema gap rather than a physics failure: Tier 1a caps `sizes_basins` at 2048 entries and, unlike `sizes_recurrent`, never stored a basin histogram, so for a unit whose basin list hit the cap the multiset is genuinely lost. Every one of the 76 failures is a truncated record and no untruncated record fails.

Asserting the check away would have hidden a real gap. Instead those units were recomputed from the rule definition and stored alongside, in `results_basins/`, this time *with* the histogram, and cross-checked against the Tier-1a fields that did survive (`n_scc`, `n_recurrent`, `shared_basin_size`, `transient_depth`).

quantity	value
units recomputed	77
largest N recomputed	21
basin sum rule holds	77/77
agrees with the surviving Tier-1a fields	77/77
recomputation time	1.17 h
truncated and NOT yet recomputed	0
Tier-1a unit ergodic: basins never computed	56
N beyond the Tier-1a sweep	25

The gap is closed: no unit remains truncated-and-unrecomputed. The checks that are still unavailable are unavailable for reasons that are not defects. Fifty-six of them are units where the Tier-1a Tarjan pass took its ergodic early exit and so never computed basins at all — there the check is *undefined*, not missing. The remaining twenty-five are the $N = 17$–18 units of the targeted extension, which lie beyond where the Tier-1a sweep ever ran; the sector quantities there are complete, only the monitored basin comparison has nothing to compare against.

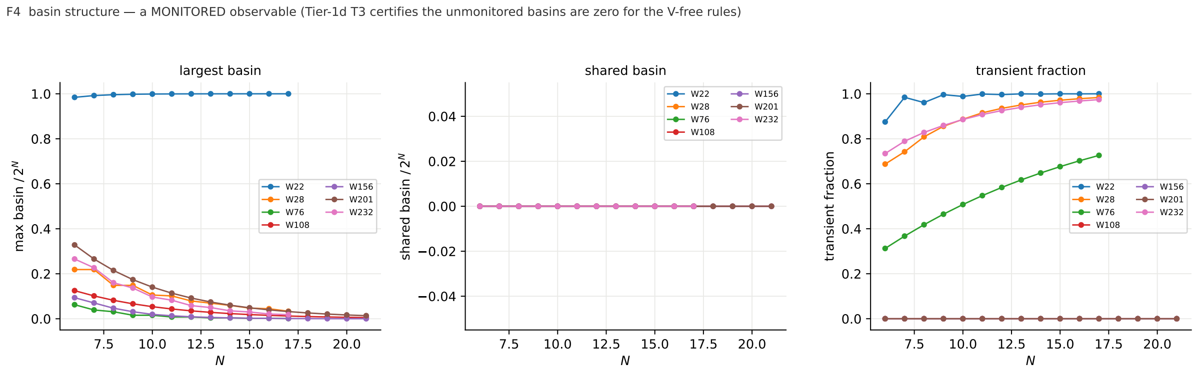

Figure 9: **F4, basin structure (MONITORED).** Largest basin, shared basin and transient fraction against N for selected rules. Two things to read off. The shared basin is *identically zero* for every rule shown: each transient state funnels to a unique terminal SCC, so the basins alone already partition the transient part and no state is ambiguous between attractors. And the two unitary rules (156, 201) sit at transient fraction 0 by construction — every state is recurrent — which is the cleanest illustration of why the transient fraction is a monitored quantity and not a sector one. Tier 1d's certificate T3 gives certified zeros for the *unmonitored* basins of the V-free rules, so for those 80 rules the monitored and channel populations coincide and these curves are channel-exact; for the rest they are monitored-only.

12 Validation

check	applicable	passing	note
A1 sum rule $\sum_k D_k = 2^N$	2892	2892	max residual 0
A2 unitary $n_{\mathrm{wcc}} = n_{\mathrm{rec}} = n_{\mathrm{scc}}$	102	102	size multisets identical
basin sum rule (MONITORED)	2892	2811	81 still unavailable
V1 hyperbola $ab \geq 2$	254	254	0 violations

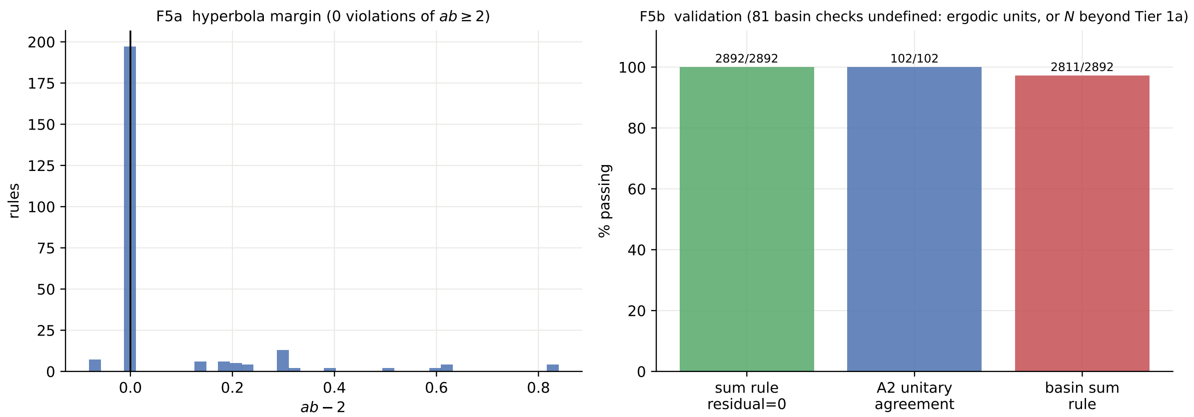

Figure 10: **F5, validation panel.** Left: distribution of the hyperbola margin $ab - 2$. Right: pass rates for the sum rule A1 (residual identically zero), the unitary agreement A2, and the monitored basin sum rule.

On the cross-tier assertions: A2 (for unitary rules, $n_{\mathrm{wcc}} = n_{\mathrm{recurrent}} = n_{\mathrm{scc}}$ with identical size multisets) holds wherever both records exist. A3 ($n_{\mathrm{wcc}} \leq n_{\mathrm{scc}}$) is structural and holds throughout. A4 is asserted only in the direction that is actually true — every sector contains at least one terminal SCC, hence $n_{\mathrm{recurrent}} \geq n_{\mathrm{wcc}}$; the reverse bound is false in general and is not asserted.

13 Comparison hooks for Tier 2

The numbers the wall grammar has to reproduce, in order of how sharply they discriminate:

1. **Rule 28's plastic number.** Equation (2) and $\rho = 1.32472$, exact. The transparency argument that predicted φ is wrong; a correct grammar must explain why one reset symbol takes $\varphi \to \rho$, i.e. why $x^2 = x + 1$ becomes $x^3 = x + 1$.

2. **The six rules with a Fibonacci sector count.** 140, 156, 196, 198, 206, 220 all have $a = \varphi$ exactly at `obc0`; the grammar should say why these six and no others, and should predict b, which the hyperbola confines to $[1.2361, 1.2369]$.

3. **The exact-base census** of §10: every rule whose sector count satisfies an integer recurrence, with the recurrence.

4. **The on-curve population.** 189 of 256 rules sit at $ab = 2$ exactly, i.e. their sector sizes are all comparable. Which symbol tables force that is a grammar question.

5. **Rule 150's family** (§9). The wall charge is shared by six dissipative rules and broken by exactly two, `IDDI` and `IEEI`. A grammar that produces the domain-wall counting should say why the same-reset-type pair is the one that breaks it.

6. **Which V+reset rules escape the collapse** (§6.1). Only 10 of 160 keep any sector structure and only 8 stay exponentially fragmented (§8.2); a grammar should predict that list, and it is a short one.

14 Open items

1. **`pbc`.** This report is `obc0` only, by instruction. The ring is a separate report; note that rule 156's Lucas law and rule 22's attractor count are `pbc` statements already verified here in passing.

2. **Separating $3^{1/4}$ from $4^{1/5}$ empirically** for the φ family. R2 §3 settles it by derivation and this report follows that; the two candidates differ by 0.26%, so confirming it from data alone would need far more than the eleven sizes available here.

3. **W_{90} and W_{165}** (both `DEED`/`EDDE`) have irregular sector counts that no growth law fits. Their structure is presumably number-theoretic in N; nothing in the engine is wrong (A1 holds exactly), but they have no place in a base plane.

4. **The five polynomial-sector rules** 74, 88, 104, 173, 233: confirm $b = 2$ directly rather than through a fit, e.g. by showing $D_{\max}/2^N$ is bounded below.

5. **A basin histogram in the Tier-1a schema**, so the gap of §11 cannot recur for future units.

15 Reproduce

```
# the sector sweep (N in the outer loop: uniform coverage if stopped early)
python -m qca_fragmentation.wcc_sweep --which all --bc obc0 \
       --n-min 6 --n-max 16 --no-stop-on-ergodic --order n

# a single rule
python -m qca_fragmentation.run_rule --rule 156 --N 6-16 --bc obc0 --tier wcc
```

```
# recompute the basin units the Tier-1a archive cannot answer
python -m qca_fragmentation.basin_recompute --bc obc0

# descriptors + hyperbola, then tables and figures F1-F5
python -m qca_fragmentation.scaling.sectors      --bc obc0
python -m qca_fragmentation.scaling.r9_tables    --bc obc0
python -m qca_fragmentation.scaling.sector_figure --bc obc0 --rebuild

python -m pytest qca_fragmentation/tests -q
```

Data: `results_tier1e/{rule}_{bc}.jsonl` (one record per unit with the full sector-size histogram, the derived observables and the A2–A4 check results), `results_basins/{rule}_{bc}.jsonl` (the independent basin recomputation), `analytics/sector_map_obc0.json` (per-rule descriptors, hyperbola verdicts and attractor deficits), `analytics/sector_validation_obc0.json`, and `checkpoints/manifest.jsonl` for the audit trail.